FBISE Physics 1st Year Most Important & Difficult Past Paper Numericals

In an experiment a 3kg water rocket is launched from ground. The rocket total energy at the top of its flight is 2352J.

a. What was the rocket launching speed?

b. What height did the rocket reach?

c. What is the kinetic energy and potential energy of the rocket after 2.5s of its launch?

Two spheres A and B are of the same masses but the radius of A is greater than that of B. Which one will require more torque applied on it to bring into rotation about their axes of rotation?

If 280 J of work is done in carrying a charge from a place where potential is -12V to another place where potential is V volt. Calculate the value of V.

Two tuning forks of frequencies 456 Hz and 452 Hz are sounded together. How many beats are produced over a period of 10 second?

A 40 kg mass is supported by 5 m long aluminium wire ($Y_A=7\times10^{10}$ Pa). To have same elongation in a copper wire of the same length under the same weight, find the diameter of copper wire. ($Y_C=12\times10^{10}$ Pa)

Find the angle of projection of a projectile for which the maximum height and corresponding range are equal?

Radius of sphere is measured with a screw gauge and is substituted in the formulae of its surface area($A=\pi r^2$) and volume($V=4/3\pi r^3$). Which of the two results will be more accurate?

Calculate the angle of projection for which K.E. at the highest point of its trajectory is equal to one-fourth of its K.E. at point of projection.

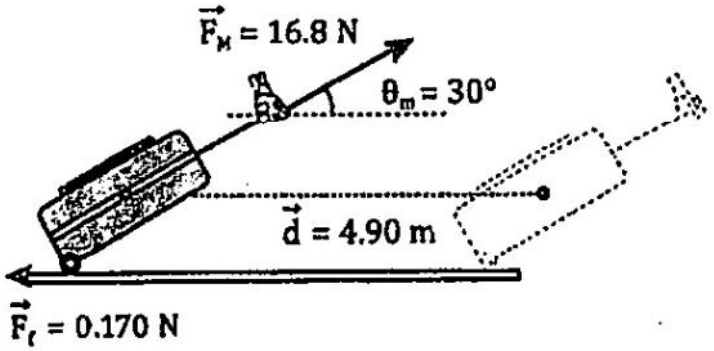
Velocity of water in 6 inch diameter pipe is $5 ft\, s^{-1}$. find the velocity in a 3 inch diameter pipe which connects with it.

A bus horn produces sound of frequency 200 Hz and is receding from a stationary observer at 18 m/s. if speed of sound is 331 m/s then what frequency does the observer hear?

A ball of mass 100 g is thrown vertically upward at a speed of $25 ms^{-1}$. If no energy is lost, determine the height it would reach.

Prove that the given equation $v_f = f_i + at$ is dimensionally correct.

A girl is dragging a suitcase and pulls it a distance of 4.90 m along level ground. She is applying a constant force of 16.8N at 30° with the horizontal. A 0.170 N friction force opposes the suitcase's motion. Find **(a)** work done by Girl **(b)** work done by frictional force and **(c)** net work done by all the forces acting on the suitcase?



An electric field has a strength of 5 N/C. What force is exerted on a charge of 2 C placed in this field?

Calculate the strain energy stored in a 1m long steel wire connected to the roof, holding a weight of 980N. The diameter of steel wire is 0.004m and $\Delta L = 0.00039m$.

State Coulomb's Law. Write its mathematical expression. Two point charges $q_1 = 4\mu C$ and $q_2 = 2\mu C$, exert a repulsive force of 36N on each other. Calculate the distance between the charges.

A peddler is pushing a trolly on a horizontal road with a force of 50 N making 30° with the road. Find the horizontal and vertical components of force?

If voltage measured across a conductor is $V \pm \Delta V = (7.3 \pm 0.3)$ volts and current, is $I \pm \Delta I = (2.73 \pm 0.051)$ ampere. Find the resistance and uncertainty in it?

At what speed would the mass of proton be tripled? The rest mass of proton is 1.673×10^{-27} kg.

A particle with charge -3e moves at a velocity of 1×10^{6} m/s at an angle of 45° and experiences a magnetic force of 6.72×10^{-14} N. What is the magnetic field strength?

Calculate the strain energy stored in a 1m long steel wire connected to the roof, holding a weight of 980N. The diameter of steel wire is 0.004m and $\Delta L = 0.00039m$.

The frequency of a certain string is 125 Hz, when it vibrates in 5 segments. Calculate the frequencies when the string is vibrating in 3 and 4 segments. What is the speed of the wave in it, if its length is 0.80m?

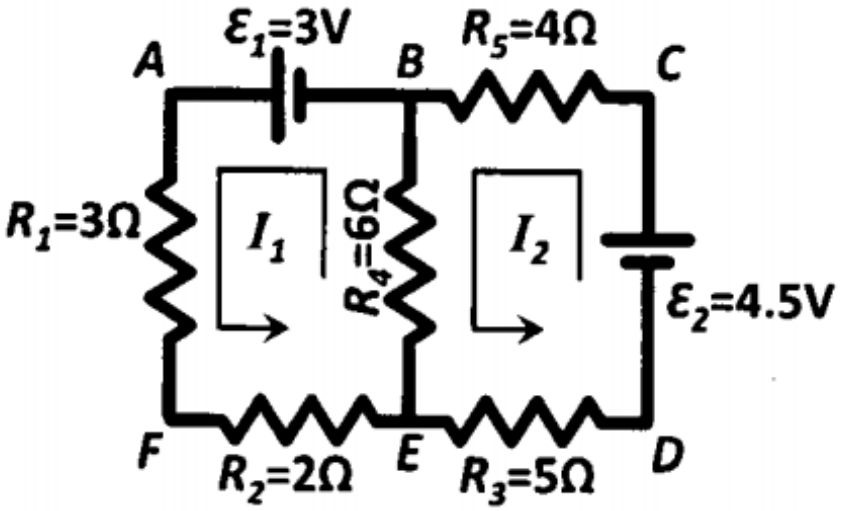
A copper wire has a length of 50 m and cross-sectional area of 1.5×10^{-6} m^2. The resistivity of copper is 1.68×10^{-8} Ω m. Calculate the resistance of the wire?

If $|A \times B| = 3/4$ and $|A . B| = 1/2$ then find the angle between **A** and **B**?

Calculate the resistance of wire 15 m long that has a diameter of 1.5 mm and resistivity of 2.63×10^{-2} Ω m.

A proton is moving in a circular orbit of radius 12 mm in a uniform 0.30 T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.

Find Current flowing through each resistor, using Kirchhoff's law, in the given circuit.



An object having a net charge of 24μC is placed in a uniform electric field of 610 N/C directed vertically. What is the mass of this object if it floats in the field?

The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C. Calculate the distance of the point from the charge.

A ball of mass 120 g is released from a height of 30 m. if the ball encounters an air resistance of 0.3 N, find the kinetic energy of the ball just before striking the ground.

Calculate the electric field intensity at a distance of 0.5 m from a point charge of 2×10^{-6} C in a vacuum.

Find the efficiency of a heat engine operating between 427 °C and 27 °C.

A current of 2 A is flowing in a copper conductor with a cross-section of 1 mm^2. Find the drift velocity of the electrons. (For copper, $n=8.5 \times 10^{28}$ per m^3)

A point source of sound emits sound energy at a rate of 15 W. Calculate the intensity of sound at a distance of 4 m from the source.

A projectile is launched horizontally from the top of a 50 m high cliff and lands at a distance of 20 m from the base of the cliff. Determine the magnitude of the launch velocity.

The energy of photon is given by E=hf, find the dimensions of Plank's constant h, where f is frequency.

Find the power delivered by an engine for attaining velocity= $(4\hat{i}+5\hat{j})$ m/s while it exerts a force= $(8\hat{i}-2\hat{j})$ N.

A rabbit jumps over an object 0.7 m high:

a. Considering just its vertical motion, calculate its vertical speed when it leaves the ground.

b. How long a time it stays in the air?

Find the power delivered by the engine for attaining velocity (2,3) m/s, while it exerts a force (7,-3) N.

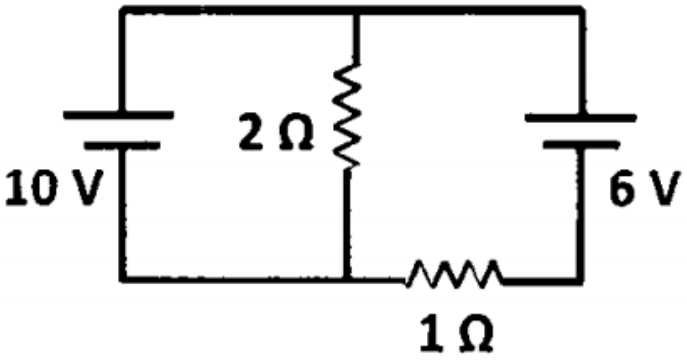
To increase intensity of waves by a factor of 25, calculate the required increase in amplitude of wave.

Calculate entropy change when 800g of ice melts at $0°C$, latent heat of fusion for ice is $3.36\times10^{5}\,J/kg$.
Also discuss sign of entropy.

A space ship has diameter of **8m** rotates about its axis to produce artificial gravity of magnitude **17.47ms^{-2}** in it. Calculate with which angular speed the satellite rotates?

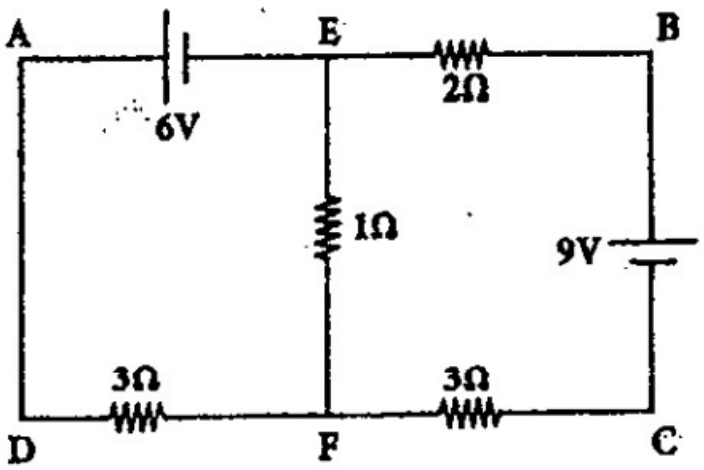
Time for 10 vibrations of a simple pendulum is 17.6s. Least count of stop watch is 0.1s. Find time period of simple pendulum. Also determine uncertainty in it.

Find the current flowing through each resistor of the circuit.



At what speed would the mass of proton be doubled? The rest mass of proton is $1.673\times10^{-27}\,kg$.

Calculate the current that flows in the 1Ω resistor in the following circuit.



The velocity of a missile traveling at 1500 ms^{-1} increases to 5000 ms^{-1} after moving through a distance of 70 km. Calculate the time for the missile to reach this velocity.

A body at 200°C undergoes a reversible isothermal process. The heat energy removed in this process is 7300 J. Calculate the entropy change and discuss entropy sign.